

DANIAL AMIN

Doctoral Student | Doctoral Researcher

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PROFILE SUMMARY

As a PhD candidate, my research combines expertise in data science, machine learning, and large language models to address stereotype leakage and bias in AI-generated user representations, particularly for marginalized communities. I develop demographically accurate and culturally grounded personas using retrieval-augmented generation (RAG) frameworks and open-source LLMs. My work emphasizes inclusive data collection, rigorous evaluation methodologies, and ethical frameworks, ensuring that AI systems authentically represent underrepresented groups and align with the principles of responsible AI.

SKILLS AND COMPETENCIES

Research Interests: AI-Generated Personas for Social Good, Bias Detection and Mitigation in AI Systems, Ethical AI for Marginalized Communities, Data-Driven Persona Creation, Generative AI and Deepfake Technologies, Cross-Cultural AI Applications, Algorithmic Fairness in Persona Development, AI Ethics and Policy Making, Inclusive Design for Global South, Quantitative and Qualitative Persona Evaluation Methods

AI & Data Science: Predictive Machine Learning, Reinforcement Learning, Deep Learning, Deep Neural Networks, RNNs & GANs, Bayesian Optimization, Generative AI, Large Language Models, Explainable AI, Retrieval Augmented Generation, Exploratory Data Analytics, Predictive and Prescriptive Analytics, Time Series Analysis & Forecasting

Programming Languages & Libraries: Python, R, Pytorch, PySpark, TensorFlow, Keras, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn, NLTK, MATLAB, LabVIEW, LangChain, LangSmith, Docker, SpaCy, Dragonfly, SQL

Software & Developer Tools: Tableau, Power BI, Looker, Minitab, SPSS, Visual Studio, PyCharm, Anaconda, JupyterLab, Docker

Cloud Architecture: Microsoft Azure, Google Cloud Platform, Amazon Web Services

Languages: English (C1), French (A1), Urdu (C2), Punjabi (C1)

EDUCATION

University of Vaasa, Finland

Doctor of Science in Business Administration and Economics

Rennes School of Business

Masters in Business Administration with Distinction (4.9/5.00)

Massachusetts Institute of Technology

Micro Masters in Statistics and Data Science

University of Wisconsin, Madison

Micro Masters in Business: Data, Insights, and Analytics

University of California San Diego

Micro Masters in Data Science

Rochester Institute of Technology

Design Thinking

Institute of Space Technology

B.S. in Aerospace Engineering

Jan, 2025 – Date

On-Campus, Finland

Sep. 2023 – Sep. 2024

On-Campus, France

Aug. 2019 – Nov. 2023

Distance Learning, USA

Aug. 2022 – May 2023

Distance Learning, USA

Aug. 2021 – Mar. 2023

Distance Learning, USA

Jan. 2022 – Nov. 2023

Distance Learning, USA

Sep. 2013 – Aug. 2017

On-Campus, Pakistan

EXPERIENCE

Research Experience

Senior AI & DS Consultant

Illusian Tech Office (Impact Lab)

May 2026 - Present

Helsinki, Finland

- Leading the generative AI evaluation framework to assess impact of generative AI on society.

Senior Data Scientist

ENI Plenitude

Nov. 2025 - Apr. 2026

Hybrid, France

- Leading the data science practice, developing solutions for the marketing, energy consumption, prediction and forecasting.
- Creating dedicated, orchestrated solutions for churn prediction due to price sensitivity.
- Developed a simulator to predict the effect of price on churn, acquisition, and expected margins.
- Developing a emotional AI analyst, capable of identifying patterns in textual communication with clients.
- Leading a team of four early to mid level practitioners and supporting them in their learning of better applied practices of data science especially in production.

AI Research Scientist

Samsung Design Innovation Center

Nov. 2024 - Nov 2025

Hybrid, France

- Developing the internal design companion for product and computational designers, combining the Generative and Predictive AI solutions to support data-driven design optimization.
- Spearheading development of multi-stages unstructured DB to act as ground truth for the LLM to enhance the data-driven design optimization process at SDIC.

AI & Data Science Lead

i13 Ventures Studio

Sep. 2023 - Nov 2024

Remote, Ireland

- Developed an LLM-based venture innovator, improving product performance by grounding and contextualizing the model through multi-query, re-ranking RAG algorithms.
- Created AI-based financial advisor, optimizing savings and investment plans by 25% through explainable AI advice using observer/observee agentic framework.
- Designed short, medium, and long-term strategy for business incubation using generative AI, enhancing incubation cycle impact by 30% with a focus on explainability.

Lead Data Scientist

Feb. 2022 - Feb. 2023

Automat Solutions Inc

Remote, USA

- Built machine learning pipelines for battery performance prediction, achieving 95% accuracy by accurately estimating the algorithmic uncertainties.
- Optimized battery material performance, increasing throughput by 40%, by using a combination of optimization algorithms.
- Automated ML pipelines, classifying chemical compositions with 90% precision, focusing on explainability and fairness.

Lead ML Engineer

Feb. 2021 - Feb. 2022

Omdena Inc

Remote, USA

- Improved image classification model accuracy by 12%, leading a team of 5 engineers to tackle recall minimization by incorporating HIL feedback.
- Developed an object recognition pipeline, enhancing detection accuracy by 15%, with a focus on reducing bias in image datasets.

Assistant Manager

Aug. 2017 - Nov. 2022

Government of Pakistan

On-site, Pakistan

- Implemented a fuel consumption estimator, increasing accuracy by 72%, by developing surrogate model.
- Developed a SMART Maintenance System, optimizing schedules for 52 aircraft components through anomaly detection algorithm in time series data.
- Introduced AI-enabled early warning systems, reducing flight risks by 27% through risk assessment models.

Management Experience

AI & Data Science Lead

Sep. 2023 - Nov. 2024

i13 Ventures Studio

Remote, Ireland

- Mentored and nurtured startups, increasing implementation of AI solutions by 50%, focusing on ethical AI and bias mitigation in emerging markets of South Africa.
- Envisioned AI and ML strategy, leading to a 15% improvement in organizational efficiency.

Head of Data Analytics — Lead Data Scientist

Feb. 2023 - Sep. 2023

Abacus Global

On-site, Pakistan

- Improved data processing speed by 40% and predictive analytics model accuracy by 25% through the implementation of a big data analytics framework.
- Led a cross-functional team to drive data-centric strategies, addressing biases and ensuring fair data practices.
- Developed solutions that improved customer satisfaction by 30% by addressing biases in data analysis.

Teaching Experience

Teaching Assistant

Aug. 2016 - Aug. 2017

Institute of Space Technology

On-site, Pakistan

- Assisted in Aero-Vehicle Performance and Design courses by providing hands-on tutorials and problem-solving exercises.
- Supported course material development, enhancing the learning experience for over 100 students by focusing on hands-on and applied content.

PUBLICATIONS

Journal Articles

- **Generative AI Personas Considered Harmful? Examining the Risks of Algorithmic User Representations in HCI Design** 2025
Amin, D., Salminen, J., Jansen, B., Shin, J., & Kim, D. H., *International Journal of Human-Computer Studies* [IJHCS](#)
- **AI Representing Personas Representing User Groups: Applying the Agency Theory to Examine Interaction Challenges of 2026 Conversational Personas as Decision-Making Tools**
Salminen, J., Jung, S.-g., Kaate, I., Xuan, T., Azem, J., Aldous, K., Amin, D., & Jansen, B. J., *Decision Support Systems* [DSS](#)
- **How to Anthropomorphize Conversational Agents in Information Systems?: A Review of Humanizing Attributes** 2025
Amin, D., Salminen, J., Fabillar, A., Sengün, S., & Jansen, B., *Online Information Review [Under Review]* [OIR](#)
- **Assessing Inter-Rater Agreement Among Large Language Model Judges: Evidence from Persona Evaluation Across Educational Risk Contexts** 2025
Amin, D., Salminen, J., & Jansen, B., *Technological Forecasting and Social Change [Under Review]* [TFSC](#)

Book Chapters

- **Large language models (LLMs) in human-computer interaction: Using LLM-generated personas to model everything from minority views to entire ecosystems** 2024
Amin, D., Salminen, J., Jansen, B., Kaate, I., & Akhtar, W., *Artificial Intelligence & Large Language Models: A Scientific Perspective*, CRC Press

- **How Can Tensions in Sustainability Transitions Be Made Visible?: Challenging the Actor-Centered Ecosystem Approach** 2026
through Persona-Based Modeling
Amin, D., Salminen, J., Ahmed, F., Zemskova, E. & Jansen, B., *Ecosystems and Networks for Sustainable Business*, Palgrave-McMillen
Conference Publications - Published
- **Creating and Evaluating Personas Using Generative AI: A Scoping Review of 81 Articles** 2026
Amin, D., Salminen, J., Ahmed, F., Tervola, M.H.S., Sethi, S., & Jansen, B., *ACM Conference on Human Factors in Computing Systems (CHI)* [CHI'26](#)
- **"Pathways to the Metaverse": Exploring the User Experience Mechanisms Driving Technology Acceptance in Virtual Lab Visits with an LLM-powered Avatar** 2026
Tu, X., Biondani, F., Amin, D., Fabillar, A., Xuan, T., Adeel, H.B., Tervola, M.H.S., Berlinger, C., Fummi, F., Salminen, J., & Jansen, B., *International Conference on Intelligent User Interfaces* [IUI'26](#)
- **"When AI Writes Personas": Analyzing Lexical Diversity in LLM-Generated Persona Descriptions** 2025
Sethi, S., Salminen, J., Amin, D., & Jansen, B., *Extended Abstracts of CHI Conference on Human Factors in Computing Systems* [CHI'25](#)
- **The Use of Large Language Models in HCI: A Critical Analysis of Synthetic Users** 2025
Salminen, J., Amin, D., Jung, S., & Jansen, B., *Augmented Humans International Conference 2025* [AH'25](#)
- **How Is AI Used for User Representation?: An Analysis of 83 Persona Prompts** 2025
Salminen, J., Amin, D., & Jansen, B., *ACS/IEEE International Conference on Computer Systems and Applications* [AICCSA'25](#)
- **Representing Religious Practices via AI-Generated Personas: A Case Study of Ramadan Behaviors from Four Predominantly Muslim Countries** 2025
Alqadi, L., Jung, S., Amin, D., Alabed, A., Salminen, J., & Jansen, B., *ACS/IEEE International Conference on Computer Systems and Applications* [AICCSA'25](#)
Conference Publications - Under Review
- **Do LLMs Know How Many Personas to Generate? A Benchmarking Study in Customer Segmentation** 2026
Amin, D., Salminen, J., & Jansen, B., *ACS/IEEE International Conference on Computer Systems and Applications [Under Review]* [AICCSA'26](#)
- **"Are These AI Personas Really Personas?": Investigating to What Extent LLM-Generated Personas Follow Persona Design Theory** 2026
Amin, D., Salminen, J., & Jansen, B., *Conference on Conversational User Interfaces [Under Review]* [CUI'26 - Poster](#)
- **Introducing Persona Ecosystem Playground (PEP): Generating Behaviorally Grounded Personas from Think-Aloud Data Using LLMs with RAG** 2026
Amin, D., Salminen, J., & Jansen, B., *Conference on Conversational User Interfaces [Under Review]* [CUI'26 - Poster](#)
- **"One Metric to Rule Them All?": Proposing A Quantitative Framework for Persona Set Diversity Measurement Toward Inclusive User Representation** 2026
Amin, D., Salminen, J., & Jansen, B., *IEEE Access [Under Review]* [IEEE Access](#)
- **How Does Anthropomorphism Affect User Outcomes?: A Meta-Analysis of Conversational Agents** 2026
Amin, D., Salminen, J., Fabillar, A., Sengün, S., & Jansen, B., *Conference on Conversational User Interfaces [Under Review]* [CUI'26](#)
- **What Principles Do Researchers Apply When Using LLMs to Create Personas?: A Review of Prompt Design for AI-Generated Personas from Fairness and Technical Perspectives** 2026
Amin, D., Salminen, J., & Jansen, B., *NordiCHI [Under Review]* [NordiCHI'26](#)
- **On the Need for Computational Resources for the Scientific Benchmarking of Data-Driven Personas** 2026
Salminen, J., Amin, D., & Jansen, B., *NordiCHI [Under Review]* [NordiCHI'26](#)
Courses
- **Bayesian Machine Learning for Optimization in Python** 2023
Educative Inc [Educative](#)

PROJECTS

PEP-Persona Ecosystem Playground

PEP

- Developing an interactive playground to support interaction of representative personas of different stakeholders among themselves and with human-in-loop
- Developing a framework to isolate user representation from the user information
- **Tech Stack** :Python, LangChain, Pinecone, Azure, LangServe, FastAPI

Bob-The Startup Advisor

- Created a selection and scoring agent to assist innovation hubs around the globe in evaluating and selecting startups in emerging markets holistically using LLMs
- Developed and deployed an enterprise level contextualized and grounded large language model for supporting startups through ideation and incubation stage.
- **Tech Stack** :Python, LangChain, Pinecone, Azure, LangServe, FastAPI

Bayesian Optimization Network for Chemical Composition Optimization

- Implemented a project for generating chemical compositions and optimizing the process using targeted DNNs through BO, Active Learning, and Meta-learning models.
- **Tech Stack** :Python, DragonFly, Bayesian Neural Networks, Genetic/Evolutionary Algorithms

AI-Shyr: Where ML meets poetry

- A pre-LLM based poetry generator for Urdu language using corpus of local poets
- **Tech Stack** :Python, Streamlit, RNNs, LSTM, nGrams,Space,NLTK

COURSES & CERTIFICATIONS

- Introduction and Advanced Topics in **Model Context Protocol** by **Anthropic** [Anthropic](#)
- Professional Certificate in **Artificial Intelligence and Machine Learning** by **European IT Certification Academy** [EITCA](#)
- Program in **Data Science and ML: Making Data-Driven Decisions** by **MIT Institute for Data, Systems, & Society** [MIT IDSS](#)
- Professional Certificate in **Data Analytics** by **Google** [Coursera](#)
- Professional Certificate in **Project Management** by **Google** [Coursera](#)
- Professional Certificate in **Data Science** by **IBM** [Coursera](#)
- Specialization in **Deep Learning** by **Deeplearning.ai** [Coursera](#)
- Specialization in **Natural Language Processing** by **Deeplearning.ai** [Coursera](#)
- Specialization in **Reinforcement Learning** by **University of Alberta** [Coursera](#)
- Complete Career track of **Data Scientist** by **365DataScience** [365DataScience](#)
- Expert track in **Data Science Fundamentals on Microsoft Azure** by **CloudSwyft and Microsoft** [FutureLearn](#)
- Nano degree in **AI Programming with Python** by **Udacity** [Udacity](#)
- Developer Certification in **Data Analysis with Python** by **FreeCodeCamp** [FreeCodeCamp](#)

HONORS & AWARDS

International

- **Climate Change Scholar** for **French Excellence Scholarship** by **Campus France** 2023-24
- **Forward Fellow** for **McKinsey Forward Program** by **McKinsey** 2022
- **Next Generation Tech Booster Scholarship** for **Data Science Track** by **Udacity** 2022
- **AWS Scholarship** for **ML Engineering Program** by **Udacity** 2022
- **Honors** for **Applied Data Science Lab** by **World Quant University** 2022
- **Future Learn Award** for **Thriving in Digital Workplace** by **University of Leeds** 2021
- **Future Learn Award** for **Solving Problem with Technology** by **University of Leeds** 2021
- **Bronze Medal** for **Poster Competition** by **Asian Science Camp** 2015

National

- **National Champion** for **Office Specialization** by **Microsoft** 2016
- **Salutatorian** for **Class of 2017** by **Institute of Space Technology** 2017
- **Cum Laude** for **Class of 2017** by **Institute of Space Technology** 2013-17
- **Outstanding Talent Award** for **Class of 2017** by **Institute of Space Technology** 2017
- **International Achievement Award** for **Class of 2017** by **Institute of Space Technology** 2016-2017

PROFESSIONAL SERVICE, PARTICIPATION & VOLUNTEERING

- Peer Reviewer for **International Journals**: International Journal of Human-Computer Studies; Expert Systems with Applications; ACM Transactions on Computer-Human Interaction; Online Information Review; Computers in Human Behavior; International Journal of Human-Computer Interaction; Climate Services 2024–26
- Peer Reviewer for **Academic Conferences**: WebSci'25 Papers; NordiCHI'26 Papers; CHI'26 Posters; CUI'26 Papers; CUI'26 Short Papers & Works-in-Progress; EICS'26 Late-Breaking Results; IUI'26 Posters; TEI'26 Work-in-Progress; Wirtschaftsinformatik'26 2025–26
- Program Committee Member, **WebSci25;NordiCHI'2026** 2025-2026
- Workshop Organizer, **CHI Workshop** 2026
- Special Recognition for reviewing, **CHI'26 Posters** 2026
- Participant, **Hi! PARIS Summer School on AI & Data for Science, Business and Society** 2024
- Facilitator, **Space Summer School at National Center of GIS & Space Applications** 2017–23
- Resource Person, **World Space Week** at Institute of Space Technology 2015–23
- Keynote Speaker, 5th–7th **International Conference on Aerospace Science and Engineering** 2017–21